NORMALIZATION

Reading: Chapter 8 (except 217-227)

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Additional material from Connolly

Normalization

- A theoretical foundation for the relational model
- Application of a series of rules that gradually improve the design
  - Minimize redundancy
  - Minimize dependency
- Objectives*
  - Free the collection of relations from undesirable insertion, update and deletion dependencies
  - Isolate data so that additions, deletions, and modifications of a field can be made in just one table and then propagated through the rest of the database

* Wikipedia

Functional Dependency

- A relationship between attributes in an entity
  - One or more attributes determine the value of another attribute
- An identifier functionally determines all the attributes of an entity
  - Shares.ID → stock code, firm name, stock price, stock quantity, stock dividend
  - If we know ID, we know the value of firm name, etc.
- Multivalued dependency
- Formulae
  - (stock dividend, stock price) → yield

yield=(dividend/price)*100

Notice the notation for functional dependency

Full Functional Dependency

- Yield is fully functionally dependent on stock dividend and stock price because both of these attributes are required to determine the value of yield
  - (stock dividend, stock price) $\rightarrow$ yield
- Determinant
  - An attribute that fully functionally determines another attribute
  - e.g., ID determines stock PE

Attribute Relationships

- One-to-one
  - Value of an attribute determines the value of another attribute and vice versa
  - $A \rightarrow B$ and $B \rightarrow A$
  - e.g.,
    - CH $\rightarrow$ Switzerland
    - Switzerland $\rightarrow$ CH

Attribute Relationships

- One-to-many
  - a value of one attribute determines the value of another attribute but **not** vice versa
  - $A \rightarrow B$
  - e.g.,
    - country name $\rightarrow$ currency unit
    - currency unit not $\rightarrow$ country name

Normal Forms

- Based on rules about relationships among the columns of a table
- Removes data redundancies that can cause update anomalies
- A classification of relations
  - 1NF
  - 2NF
  - 3NF
  - BCNF
  - 4NF
  - 5NF

Usually, only the first 3 normal forms are applied to a DB

Data Redundancy

- Major aim of relational database design is to group columns into tables to:
  1. minimize data redundancy and
  2. reduce file storage space required by implemented base tables
- Problems associated with data redundancy are illustrated by

Problems associated with data redundancy are illustrated in the example on the following slides

StaffDistributionCenters Table

- Note the details of a distribution center are repeated for every employee (not normal form)

StaffDistributionCenter

ID	StaffNo	staffName	position	salary	dCenterNo	dAddress	dTelNo
1	S1500	Tom Daniels	Manager	$48,000.00	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618
2	S0003	Sally Adams	Assistant	$30,000.00	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618
3	S0010	Mary Martinez	Manager	$51,000.00	D0002	City Center Plaza, Seattle, WA 98122	206-5556756
4	S3250	Robert Chin	Assistant	$33,000.00	D0002	City Center Plaza, Seattle, WA 98122	206-5556756
5	S2250	Sally Stern	Manager	$48,000.00	D0004	2 W. El Camino, San Francisco, CA 94087	822-555-3131
6	S0415	Art Peters	Manager	$42,000.00	D0003	14 - 8th Avenue, New York, NY 10012	212-371-3000

Update Anomalies

- Tables that contain redundant information may potentially suffer from update anomalies
- Types of update anomalies include:
  - Insertion – how do you insert details of a new distribution center that has no employees?
  - Deletion – when we delete the last employee in a distribution center, we lose the information about the distribution center
  - Modification – changes to a distribution center must be made for all records containing that distribution center

Better Design

StaffDistributionCenter | **Staff**

ID	staffNo	staffName	position	salary	dCenterID
1	S1500	Tom Daniels	Manager	$48,000.00	1
2	S0003	Sally Adams	Assistant	$30,000.00	1
3	S0010	Mary Martinez	Manager	$51,000.00	2
4	S3250	Robert Chin	Assistant	$33,000.00	2
5	S2250	Sally Stern	Manager	$48,000.00	4
6	S0415	Art Peters	Manager	$42,000.00	3

Staff | **DistributionCenters**

ID	dCenterNo	dAddress	dTelNo
1	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618
2	D0002	City Center Plaza, Seattle, WA 98122	206-5556756
3	D0003	14 - 8th Avenue, New York, NY 10012	212-371-3000
4	D0004	2 W. El Camino, San Francisco, CA 94087	822-555-3131

First Normal Form (1NF)

- All rows must have the same number of columns
- Single valued attributes only

No universal agreement as to what would disqualify a table from being in 1NF

Customer

Customer ID	First Name	Surname	Telephone Number
123	Robert	Ingram	555-861-2025
456	Jane	Wright	555-403-1659 555-776-4100
789	Maria	Fernandez	555-808-9633

Typical violation of 1NF

Resist the temptation to include repeated fields as CSV text

Example – Table not 1NF

Staff | DistributionCenters

ID	dCenterNo	dAddress	dTelNo
1	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618, 503-555-3619
2	D0002	City Center Plaza, Seattle, WA 98122	206-5556756
3	D0003	14 - 8th Avenue, New York, NY 10012	212-371-3000
4	D0004	2 W. El Camino, San Francisco, CA 94087	822-555-3131

Repeated field

Converting to 1NF

DistributionCenters

ID	dCenterNo	dAddress	dTelNo
1	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618, 503-555-3619
2	D0002	City Center Plaza, Seattle, WA 98122	206-5556756
3	D0003	14 - 8th Avenue, New York, NY 10012	212-371-3000
4	D0004	2 W. El Camino, San Francisco, CA 94087	822-555-3131

Replace a repeating group with a foreign key relationship

DistributionCenters1NF

ID	dCenterNo	dAddress
1	D0001	8 Jefferson Way, Portland, OR 97201
2	D0002	City Center Plaza, Seattle, WA 98122
3	D0003	14 - 8th Avenue, New York, NY 10012
4	D0004	2 W. El Camino, San Francisco, CA 94087

DistribtionCentersTelephones

ID	dCenterID	dTelNo
1	1	503-555-3618
2	1	503-555-3619
3	2	206-5556756
4	3	212-371-3000
5	4	822-555-3131

Are We on Track?

- Using Access, put the following Dentist DB in 1NF

staffNo	dentistName	patNo	patName	Appointment		surgeryNo
S1011	Tony Smith	P100	Gillian White	12-Sep-08	10.00	S15
S1011	Tony Smith	P105	Jill Bell	12-Sep-08	12.00	S15
S1024	Helen Pearson	P108	Ian MacKay	12-Sep-08	10.00	S10
S1024	Helen Pearson	P108	Ian MacKay	14-Sep-08	14.00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Sep-08	16.30	S15
S1032	Robin Plevin	P110	John Walker	15-Sep-08	18.00	S13

For now, you can consider the full name to be a single valued attribute

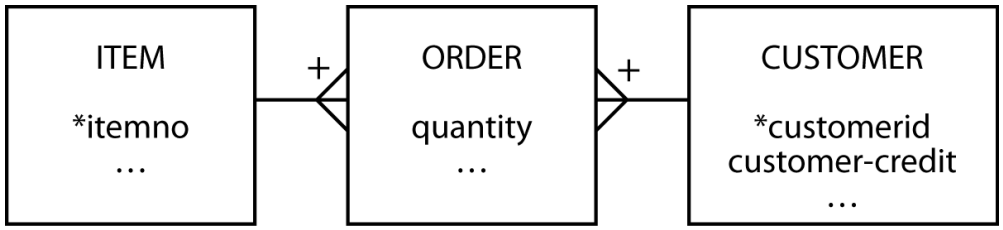
Second Normal Form (2NF)

- Violated when a non-key column is a fact about part of the primary key
- A column is not fully functionally dependent on the primary key
  - `customer-credit` in this case

Mainly applies to tables with multiple natural keys

order

itemno	customerid	quantity	customer-credit
12	57	25	OK
34	679	3	POOR

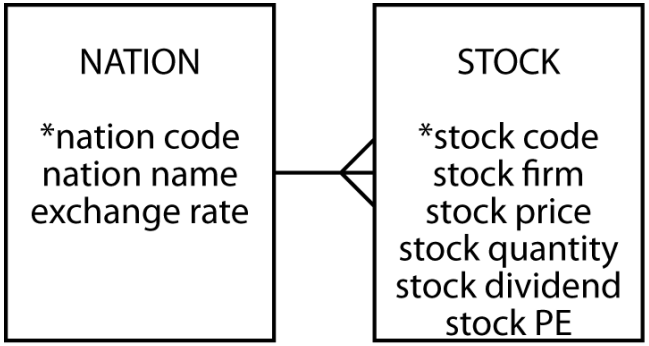


Third Normal Form (3NF)

- Violated when a non-key column is a fact about another non-key column, restated as
  - A column is not fully functionally dependent on the primary key

Exchange rate is a fact about a nation

stock		
stock code	nation	exchange rate
MG	USA	0.67
IR	AUS	0.46



Example - not 3NF

Values in dAddress and dTelNo can be determined from dCenterNo

Values in staffNo, staffName, position and salary are determined from ID

StaffDistributionCenter

ID	StaffNo	staffName	position	salary	dCenterNo	dAddress	dTelNo
1	S1500	Tom Daniels	Manager	$48,000.00	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618
2	S0003	Sally Adams	Assistant	$30,000.00	D0001	8 Jefferson Way, Portland, OR 97201	503-555-3618
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Are We on Track?

- Put the Dentist DB (slide 17) in 3NF

Are We on Track?

- Using Access, convert the DB below to a normalized (3NF) Access DB

eNum	eName	DOB	org	pos	start	finish
0056	Kate Brown	12/12/76	Pat&Co	IT Assistant	11/1/90	4/30/95
0057	Gil Bradford	3/1/71	Will&Son	Technician	6/1/88	3/31/91
0057	Gil Bradford	3/1/71	Will&Son	Asst. Researcher	4/1/91	3/31/94
0057	Gil Bradford	3/1/71	Will&Son	Project Manager	4/1/94	1/31/96
0089	Steve Widden	9/9/83	Pat&Co	IT Assistant	3/1/91	8/31/95